Ann Von Holle

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Epidemiologist & Biostatistician | HEOR, Real-World Evidence & Evidence Synthesis

Epidemiologist and biostatistician (Ph.D., Epidemiology; M.S., Biostatistics) experienced in independently designing and executing statistical analyses for healthcare, clinical, and real-world evidence (RWE) research across academia, government, and the healthcare and life-sciences industry. Expertise spans statistical analysis plans, survival and longitudinal analyses, meta-analysis and pooled analyses, and decision analytic and simulation modeling of health and economic outcomes using claims, cohort, and clinical trial data. Advanced R programmer (also SAS) with a reproducible, version-controlled workflow. Author of numerous peer-reviewed publications, including lead-author work in leading epidemiology journals. Skilled at communicating complex methods to multidisciplinary teams, sponsors, and non-technical audiences, and at mentoring junior analysts.

CORE COMPETENCIES

- **Statistical methods:** statistical analysis plans; survival/time-to-event analysis; meta-analysis and pooled multi-cohort analysis; longitudinal mixed-effects and latent growth models; structural equation modeling; regression and predictive modeling; machine learning and clustering; simulation
- **HEOR & real-world evidence:** RWE study design using claims and other real-world data; decision analytic modeling of health and economic outcomes; health technology assessment (HTA) of diagnostic tests; evidence synthesis, literature and systematic reviews; observational and epidemiologic study design
- **Programming:** R (advanced), SAS, Git, R Markdown, LaTeX, high-performance cluster computing
- **Communication & leadership:** peer-reviewed manuscripts, white papers, conference presentations, sponsor-facing deliverables, data visualization, mentoring

PROFESSIONAL EXPERIENCE

American Medical Group Association, Remote **February 2025 – Present**

Senior Population Health Research Analyst

Scope: Lead real-world evidence (RWE) research within the AMGA Research and Analytics group, covering the full research lifecycle from study concept and protocol through statistical analysis, peer-reviewed publication, and dissemination.

- Develop study concepts, research protocols, and statistical analysis plans for RWE studies addressing population health priorities.
- Design and conduct analyses of real-world data using advanced epidemiologic and statistical methods in R, with results published in peer-reviewed journals (e.g., *Journal of Managed Care & Specialty Pharmacy*, 2026; *Frontiers in Endocrinology*, 2026).
- Translate complex statistical findings into clear data visualizations, reports, manuscripts, and conference presentations for medical groups and integrated health systems across the US.
- Collaborate with sponsors, clinical subject matter experts, programmers, and analysts to scope analytic approaches and deliver projects successfully.

Optum Genomics, Remote **April 2023 – July 2024**

Associate Research Director

Scope: Conducted health economics and outcomes research, combining RWE studies with decision analytic modeling of health and economic outcomes, to assess the clinical and economic value of genetic diagnostic tests and cancer early detection (health technology assessment).

- Planned, designed, and executed a simulation study of carrier screening, building decision analytic models in R that integrated clinical validity measures and claims data to estimate health and economic outcomes.

- Analyzed medical claims data in R to evaluate the analytical and clinical validity of molecular diagnostic tests for cancer early detection.
- Conducted comprehensive literature reviews and evidence synthesis on the quality of molecular diagnostic tests to inform research and practice.
- Co-authored white papers on the analytical and clinical validity of genetic and molecular lab-developed tests, providing expert recommendations.

National Institute of Environmental Health Sciences (NIEHS), Research Triangle Park, NC

Special Volunteer **April 2023 – April 2026**

Completed the manuscript submission process for research conducted during postdoctoral fellowship.

Postdoctoral Fellow **July 2018 – April 2023**

Biostatistics and Computational Biology Branch (Mentor: Clarice R. Weinberg)

Scope: Conducted observational research on breast cancer outcomes and serum iron biomarkers: developed research concepts, specified statistical analysis plans, performed formal analyses, and authored manuscripts.

- Independently developed hypotheses and statistical analysis plans and selected data sources and analytic methods for eight lead-author manuscripts.
- Applied time-to-event (survival) analysis, longitudinal mixed-effects models, meta-analysis, and machine learning, including a pooled multi-cohort analysis with the Premenopausal Breast Cancer Collaborative Group.
- Performed large-scale analyses in a high-performance computing environment.
- Authored manuscripts and presented findings internally and at national conferences (Society for Epidemiologic Research, American Association for Cancer Research).
- Mentored graduate and undergraduate students in NIH training programs on developing analysis plans, conducting analyses, and presenting results.

University of North Carolina at Chapel Hill, Chapel Hill, NC

Graduate Researcher **August 2013 – June 2018**

Department of Epidemiology (Advisor: Kari E. North)

Scope: Doctoral research in a genetic epidemiology training program on infant growth, adolescent lipid levels, sociodemographic factors, and genetic risk variants.

- Applied latent growth mixed-effects models, factor analysis, and non-linear random-effects models to longitudinal cohort data, producing three lead-author manuscripts.
- Wrote and was awarded an American Heart Association predoctoral fellowship, “Infant Growth Trajectories and Dyslipidemia in Adolescence” (16PRE29200008).
- Documented all analyses reproducibly in R, Markdown, and LaTeX with version control in Git.

Biostatistician **September 2005 – September 2014**

Department of Psychiatry, Center of Excellence for Eating Disorders

- Planned and interpreted statistical analyses in collaboration with faculty, postdoctoral fellows, and students.
- Analyzed observational, clinical trial, and medical record data using regression and structural equation models in SAS and R, contributing to 40+ peer-reviewed publications.
- Authored statistical methods sections, created data visualizations, and ensured data integrity for publications.

EDUCATION

Ph.D., Epidemiology, University of North Carolina at Chapel Hill | *Thesis:* Infant Growth Trajectories and Lipid Levels in Adolescence

M.S., Biostatistics, University of North Carolina at Chapel Hill

B.A., Applied Mathematics, University of California, San Diego

RECENT SELECT PUBLICATIONS

*Indicates a mentored student's work.

- **Inflammatory marker profiles vary by BMI across Crohn's disease, ulcerative colitis, psoriasis, and psoriatic arthritis.**

  Shields SA, **Von Holle A**, Padival R, Merola JF, Zuckerman R, Mohl JT, Ciemins EL. *Frontiers in Endocrinology*. 2026;17:1851905. doi:10.3389/fendo.2026.1851905

- **Primary adherence to medications for the treatment of major depressive disorder: A descriptive study using real-world data.**

  Von Holle A, Mohl JT, Zuckerman RB, Obenrader J, Ciemins EL. *Journal of Managed Care & Specialty Pharmacy*. 2026;32(7):793–806. doi:10.18553/jmcp.2026.32.7.793

- **Sedentary time and breast cancer risk in the Sister Study cohort.**

  Donzella SM, Rogers M, O'Brien KM, **Von Holle A**, Sandler DP, Weinberg CR, Díaz Santana MV. *Cancer Epidemiology, Biomarkers & Prevention*. 2025;34(11):1955–62. doi:10.1158/1055-9965.EPI-25-0510

- **Association between body iron status and biological aging.**

  Von Holle A, Ramamurthy S*, Díaz Santana MV, et al. *Nutrients*. 2025;17(9):1409. doi:10.3390/nu17091409

- **BMI and breast cancer risk around age at menopause.**

  Von Holle A, Adami HO, Baglietto L, et al. *Cancer Epidemiology*. 2024;89:102545. doi:10.1016/j.canep.2024.102545

- **Infant growth trajectories and lipid levels in adolescence: Evidence from a Chilean infancy cohort.**

  Von Holle A, North KE, Gahagan S, Blanco E, Burrows R, Lozoff B, Howard AG, Justice AE, Graff M, Voruganti S. *American Journal of Epidemiology*. 2022:kwac057. doi:10.1093/aje/kwac057

- **Does this adult patient have hypertension? The Rational Clinical Examination systematic review.**

  Viera AJ, Yano Y, Lin FC, Simel DL, Yun J, Dave G, **Von Holle A**, Viera LA, Shimbo D, Hardy ST, Donahue KE, Hinderliter A, Voisin CE, Jonas DE. *JAMA*. 2021;326:339–47. doi:10.1001/jama.2021.4533

- **Evidence for familial clustering in breast cancer age of onset.**

  Von Holle A, O'Brien KM, Sandler DP, Weinberg CR. *International Journal of Epidemiology*. 2021;50:97–104. doi:10.1093/ije/dyaa201

Full publication list: https://www.ncbi.nlm.nih.gov/myncbi/ann.von%20holle.1/bibliography/public/

In preparation: Von Holle A, O'Brien KM, Sandler DP, Weinberg CR. Associations between male-origin microchimerism and breast, ovarian, and uterine cancer. (2026)

GRANTS AND AWARDS

Individual Predoctoral Fellowship (16PRE29200008), American Heart Association ($51,900) 2016 – 2018
Intramural Paper of the Month, *Environmental Factor*, NIEHS Newsletter, for "Evidence for familial clustering in breast cancer age of onset" May 2021

PRESENTATIONS

*Represents trainee's work for which Ann Von Holle was a mentor.

Oral

Ramamurthy S*, **Von Holle A**, et al. Association between body iron status and biological aging. NIEHS Scholars Connect Spring Symposium, Durham, NC. 2023
Von Holle A, North KE, Gahagan S, Blanco E, et al. Infant growth trajectories and lipid levels in adolescence. American Heart Association EPI/Lifestyle Sessions, New Orleans, LA. 2018

Poster

Von Holle A, Thompson RL*, O'Brien KM, Sandler DP, Weinberg CR. Associations between serum iron biomarkers and breast cancer tumor size. American Association for Cancer Research Annual Meeting, Orlando, FL. 2023

Ramamurthy S*, **Von Holle A**, et al. Association between body iron status and the aging process. NIEHS Scholars Connect Poster Sessions (virtual). 2022

Von Holle A, O'Brien KM, Sandler DP, Schoemaker M, Swerdlow A, Nichols HB, Jones ME, Janicek R, The Premenopausal Breast Cancer Collaborative Group, Weinberg CR. Characterizing the association between BMI and breast cancer risk by menopause status and age-time. Society for Epidemiologic Research, Chicago, IL. 2022

TEACHING AND MENTORING

NIEHS, Research Triangle Park, NC

Mentor, 1 undergraduate student, NIEHS Scholars Connect Program 2022 – 2023

Mentor, 2 graduate students, Summer Internship Program 2021, 2022

Guest lecturer, "Introduction to Statistics and Experimental Design" January 2020

Co-lead, "Mentoring of Junior Faculty and Fellows" session, Biostatistics and Computational Biology Branch retreat October 2019

University of North Carolina at Chapel Hill

Teaching Assistant, Advanced Epidemiologic Methods (https://github.com/UNC-EPID-722/2016) Spring 2016

PROFESSIONAL SERVICE AND DEVELOPMENT

Journal peer review: BMJ Open | International Journal of Cancer | Cancer Epidemiology, Biomarkers & Prevention | BMC Public Health | Epidemiology | BMC Pregnancy and Childbirth | Journal of Eating Disorders | Nutrition and Metabolism | Cancer Medicine | Clinical Epigenetics

Society for Epidemiologic Research: Annual meeting abstract reviewer 2021 – 2022

Symposium co-chair, "Polygenic Risk Scores: challenges and opportunities for population health equity" December 2020

NIEHS: Internal manuscript peer reviewer 2018 – 2022

Training: Research Mentor Training, 4-week course, NIEHS May 2019, May 2021

"Write Winning NIH Grant Proposals" workshop, NIEHS February 2019

Professional associations: Society for Epidemiologic Research | American Association for Cancer Research